Bhargavi Poyekar

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Core Competencies

- **AWS Certified Developer Associate**, **AWS Cloud Practitioner**, and **Terraform Associate (003).**
- **Cloud Engineering** - Building and deploying backend applications and data intensive services across GCP and AWS.
- **Infrastructure** - Designing reliable automation for complex workflows, infrastructure, and development environments.
- **Data & AI Engineering** - Applying data engineering and LLM-based approaches to financial data processing.
- **Systems Engineering** - Solving problems around scalability, performance, reliability, and maintainability.

Professional Experience

Associate Software Engineer, K2Q Capital Pvt Ltd. **Aug 2025 – Present**

- Refactored and scaled a distributed financial data pipeline, reducing daily processing from **approximately 1.5 hours of manual work to 8 minutes** using **Python**, **OOP**, **Cloud Run**, **Cloud Scheduler**, and **Pub/Sub**, implemented a **strategy based architecture** for evolving data sources with robust error handling and monitoring.
- Designed and developed a **Dash** financial analytics platform on GCP using **Cloud Run**, **Cloud Storage**, **DuckDB**, and **Memorystore (Redis)**, optimizing access to **2-3 GB datasets** through efficient querying and caching to reduce dashboard loading latency by **70-75 seconds**.
- Automated an equity-data pipeline, reducing weekly manual processing from **1 hour to 2 minutes** through daily scheduling; integrated **LangGraph-based LLM validation** and **web search** to automatically verify newly added companies and improve data reliability.
- Built a large-scale **LLM-powered 10-K processing pipeline** handling **around 250K SEC filings**, automating retrieval, Business and Risk Factors extraction, and generation of year-wise **Parquet datasets** for downstream analysis and data-quality assessment.

Software Developer, University of Maryland, Baltimore County **June 2024 – June 2025**

- Built a self-service lab deployment portal using **Django and Python**, enabling students and instructors to securely launch and manage multi VM environments with isolated networks for course projects saving **30+** instructor hours per semester.
- Provisioned **AWS EC2 instances** within VPC public/private subnets using **Terraform**, with secure SSH access configured via a Bastion Host and automated VM configuration managed using **Ansible**.
- Streamlined platform scalability and reusability by containerizing Node services and modularizing infrastructure using **Infrastructure-as-Code (Terraform)**, reducing deployment time by 60%.

Software Engineer (Cloud) Intern, University of Maryland, Baltimore County **Jun 2023 – May 2024**

- Automated configuration of **50 Linux VMs** using **Ansible**, standardizing system settings, documentation, tools, and permissions to provide consistent environments for **100+ students and faculty** across course exercises.
- Designed and deployed **Capture the Flag (CTF)** exercises across the VM infrastructure, automating challenge setup, configuration, and reset using **Ansible**.

Projects

Redis Compatible Database Server (Storage Engine, RESP Protocol, Persistence, Pub/Sub):

- Built a **Redis clone** from scratch using Python AsyncIO, implemented **RESP protocol**, TCP networking, and a modular **storage engine** with Strings, Lists, Sets, TTL expiration, and LRU eviction (O(1) memory tracking).
- Designed an **AOF persistence layer** with background asynchronous writes, automatic log compaction (62% size reduction), and crash recovery, achieving **durability without sacrificing throughput (12.5K req/sec)**.
- Added **real-time Pub/Sub** with concurrent subscriber management, **comprehensive observability** (10+ metrics via INFO command), and **Docker** containerization with multi-stage Alpine build (image size ¡ 100MB).
- **Tech: Python, AsyncIO, TCP Sockets, AOF, Pub/Sub, Docker, pytest.**

HDFS to S3 Secure Data Transfer Application (Python, Hadoop, AWS):

- Developed a **Python** application to automate file transfers from **Hadoop HDFS** to **AWS S3**, reducing manual data handling time by 50%. Implemented **IAM Role policies, STS AssumeRole for temporary security credentials, and client side encryption**. Enhanced security by using **AWS SSM Parameter Store** to store sensitive credentials, replacing environment variables.
- Integrated **Amazon CloudWatch** to monitor file transfers, track errors, and log performance metrics. Configured **Amazon SNS** to send email notifications for transfer status updates (success/failure).
- **Tech: HDFS, S3 Bucket, IAM Roles, Python, boto3, GIT, Linux, CloudWatch, SNS, STS, SSM.**

Technical Skills

Cloud & Infrastructure: AWS, Terraform, Docker, Ansible, GCP (Cloud Run, Pub/Sub, Cloud Scheduler, Memorystore).
Languages and Databases: **Python**, C, PHP, Go, JavaScript, SQL, SQLite, DuckDB, Redis
Frameworks & Libraries: **Django, Django REST**, Flask, FastAPI, Selenium, Pandas, NumPy, Scikit-learn, LangGraph.
Tools & Practices: **Git**, Linux, pytest, Postman, GitHub Actions, VSCode, CI/CD, REST APIs, IaC.

Education

University of Maryland, Baltimore County - Baltimore, MD | MS in Computer Science | GPA: 3.85/4.0 Aug 2022 - May 2024
Sardar Patel Institute of Technology, Mumbai, India | B.Tech in Computer Engineering | GPA: 9.4/10 Aug 2018 - May 2022